

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the subcommand to read strings in as factors.

```
#install.packages('tidyverse')
#install.packages('lubridate')
#install.packages('here')

library(tidyverse)
library(lubridate)
library(here)

getwd()
```

```
## [1] "/home/guest/GITClone/EDE_Fall2025"
```

```
here()
```

```
## [1] "/home/guest/GITClone/EDE_Fall2025"
```

```
Neonics <-read.csv(  
  file = here("./Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv"),  
  stringsAsFactors = TRUE  
)  
  
Litter <-read.csv(  
  file = here("./Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv"),  
  stringsAsFactors = TRUE  
)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: We may be interested in the ecotoxicology of neonicotinoids on insects because insects serve several important ecological roles. Insects are important pollinators and serve as the basis of most food chains. Adverse effects of neonicotinoids on insects will likely have unintended consequences for plants and animals that rely on insects for food or pollination. According to a quick search, neonicotinoids often harm and kill insects that are not the intended targets of the insecticide. Neonicotinoids are water soluble and absorbed easily, prompting them to be spread much further than where they were applied.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Studying litter and woody debris on forest floors is important because it provides habitat for several species and is integral in nutrient recycling. For land managers, another concern associated with leaf litter and woody debris is its potential of creating/worsening forest fire conditions. Thus, it is important that the Niwot Ridge LTER station has a complete understanding of ecosystem function.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Sampling sites are chosen randomly so long as they occur at a tower plot and contain woody vegetation >2 meters tall. Depending on proximity to other sites and geographical features, the number of plots fluctuates. 2. Ground traps are placed once per year at both

targeted and randomized sites within a given plot, depending on the ground cover (greater than 50% woody vegetation within a plot results in a randomized litter trap, less than 50% woody vegetation may require a more targeted trap). 3. At deciduous forest sites, sampling occurs once every two weeks during senescence. At evergreen forest sites, sampling occurs once every two months on a year-round basis.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
summary(Neonics$Effect)
```

```
##      Accumulation      Avoidance      Behavior      Biochemistry
##           12           102           360           11
##      Cell(s)      Development      Enzyme(s)      Feeding behavior
##           9           136           62           255
##      Genetics      Growth      Histology      Hormone(s)
##          82           38           5           1
##      Immunological      Intoxication      Morphology      Mortality
##          16           12           22           1493
##      Physiology      Population      Reproduction
##           7           1803           197
```

```
sort(table(Neonics$Effect), decreasing = TRUE)
)
```

```
##
##      Population      Mortality      Behavior      Feeding behavior
##          1803          1493          360          255
##      Reproduction      Development      Avoidance      Genetics
##          197          136          102          82
##      Enzyme(s)      Growth      Morphology      Immunological
##          62          38          22          16
##      Accumulation      Intoxication      Biochemistry      Cell(s)
##          12          12          11          9
##      Physiology      Histology      Hormone(s)
##           7           5           1
```

Answer: Population, Mortality, and Behavior are the most common effects that are studied. I imagine these are of interest in the Neonics dataset because researchers can track population changes of certain species due to introduction of neonicotinoids, as well as if mortality rates or the behavior of the studied species change.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name, maxsum = 5)
```

```
##           Honey Bee           Parasitic Wasp Buff Tailed Bumblebee
##           667           285           183
## Carniolan Honey Bee           (Other)
##           152           3336
```

```
head(sort(table(Neonics$Species.Common.Name), decreasing = TRUE), 5)
```

```
##
##           Honey Bee           Parasitic Wasp Buff Tailed Bumblebee
##           667           285           183
## Carniolan Honey Bee           Bumble Bee
##           152           140
```

Answer: Honey Bees, Parasitic Wasps, Buff Tailed Bumblebees, Carniolan Honey Bees, Bumble Bees, and Italian Honeybees are the six most commonly studied species in the dataset. These species are all important pollinators, which is why they are of special interest over other insect species. Researchers want to know what effect neonicotinoids have on these integral species.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

```
view(Neonics)
```

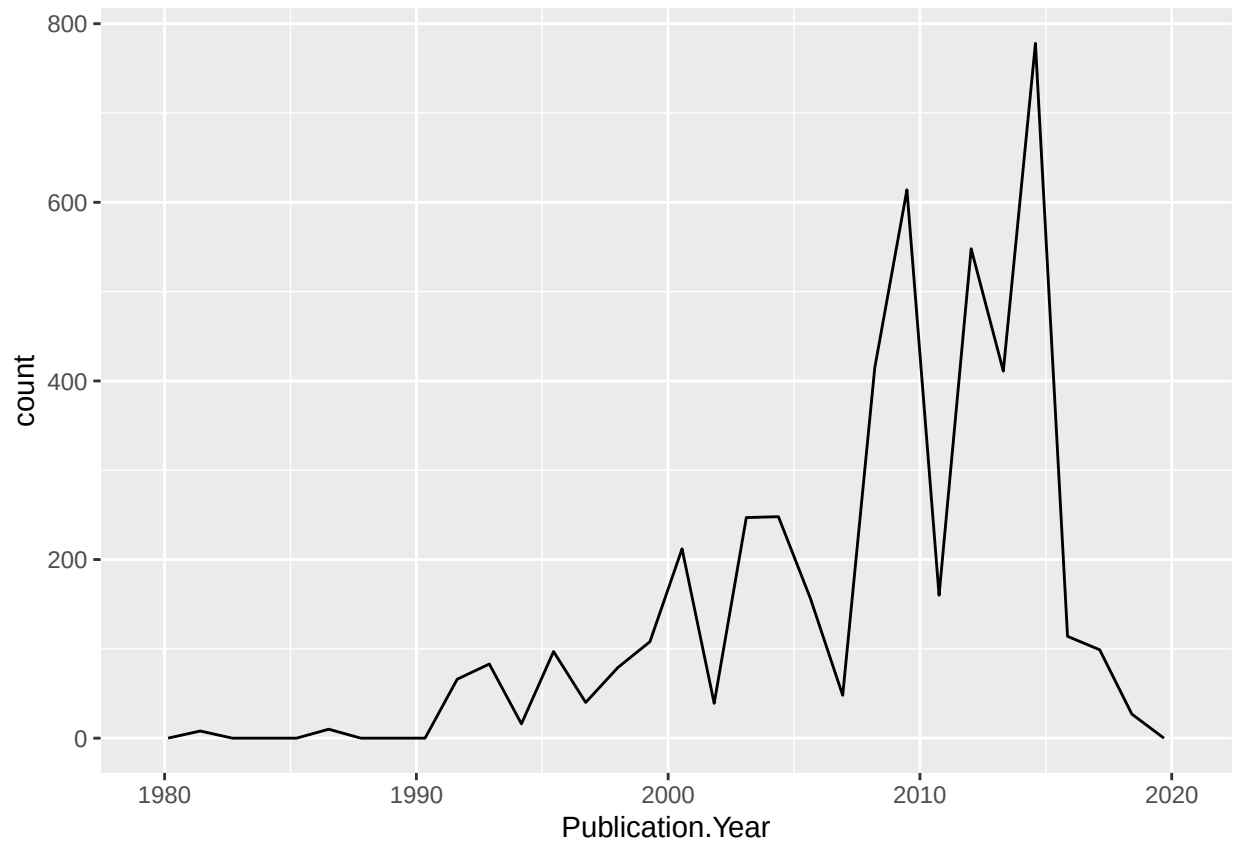
Answer: The 'Conc.1..Author.' column is classified as factor data. It is likely not numeric because some data concentrations are approximated (~), which contain symbols, and the values are also measured in different units. Thus, for comparison purposes, it would be inaccurate to compare these values numerically without converting them to a unifying metric.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
ggplot(data = Neonics,
       aes(Publication.Year)) +
  geom_freqpoly()
```

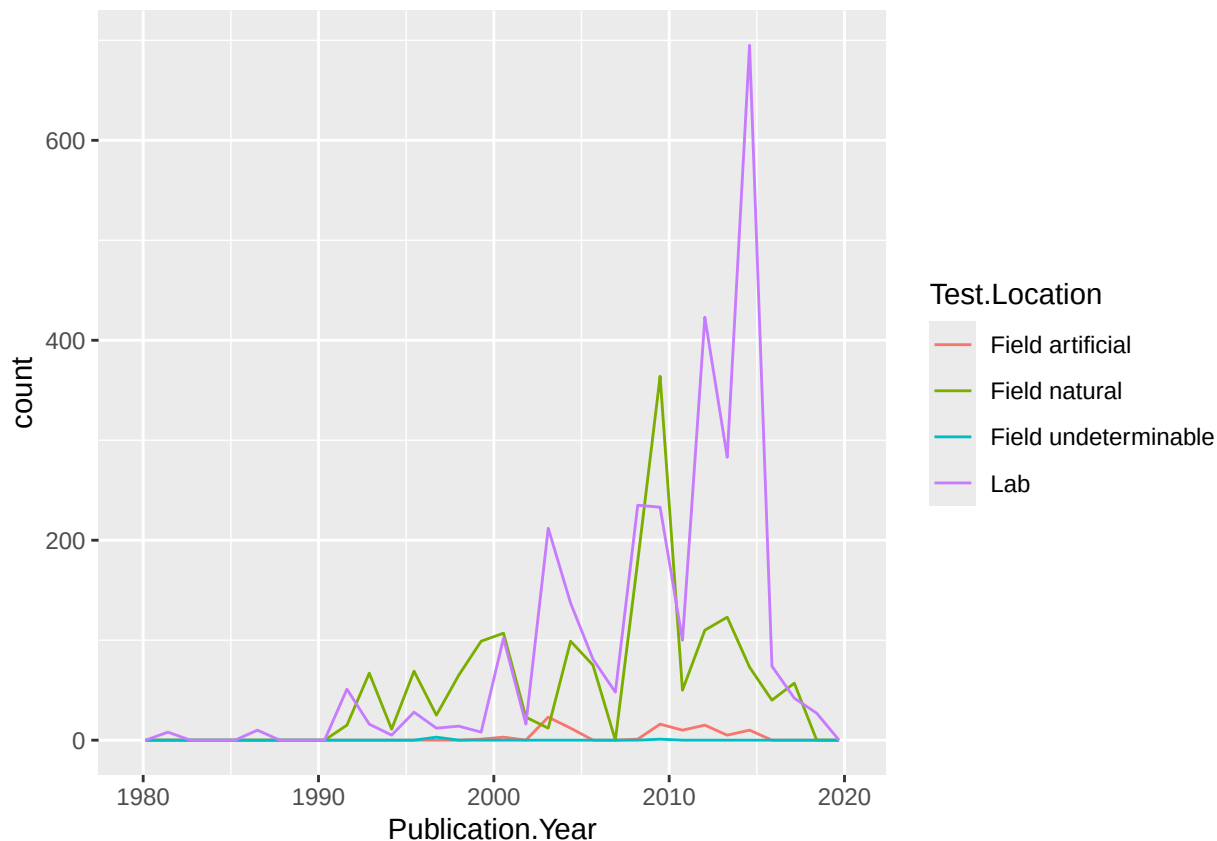
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(data = Neonics,  
       aes(x = Publication.Year,  
           color = Test.Location)) +  
  geom_freqpoly()
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



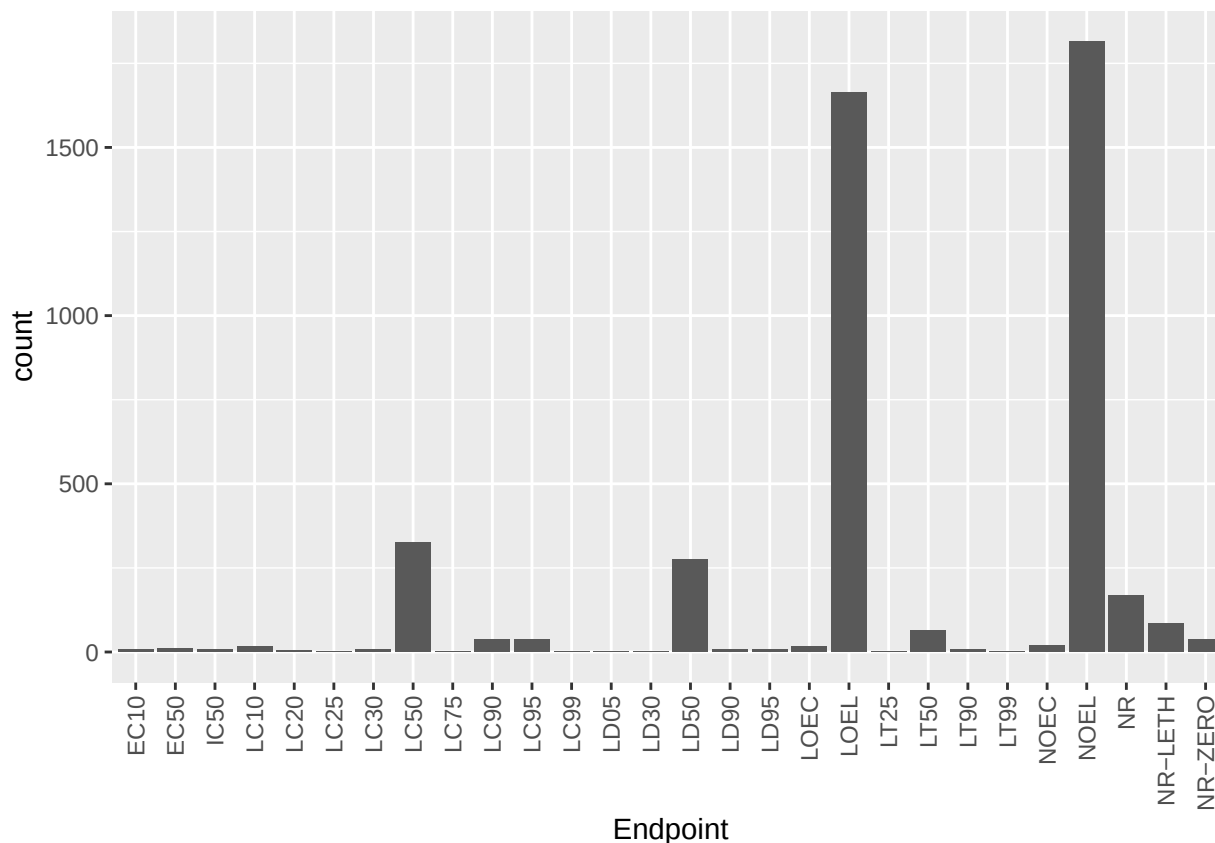
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test location is in a laboratory setting. In the 1990s, testing in the field in a natural environment was the most common. By the 2000s, laboratory tests were more favored than natural field sites but both grew in popularity. In the 2000s, there appears to be a slight interest in artificially simulating field conditions for tests, though not nearly as many of these types of studies as with lab or natural field sites. By the 2010s, lab tests shot up, tripling in number of tests. By the end of the 2010s, publications with lab tests and natural field tests plummeted to zero.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot( data = Neonics,
        aes(Endpoint)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The two most common endpoints are NOEL and LOEL. NOEL is defined as “No-observable-effect-level” meaning that even the highest dosage tested had no significant effects relative to the control. LOEL stands for “lowest-observable-effect-level”, which means that the lowest dose tested had significantly different effects on the subject relative to the control.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
Litter$collectDate <- as.Date(Litter$collectDate)
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
unique(Litter$collectDate, incomparables = FALSE)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

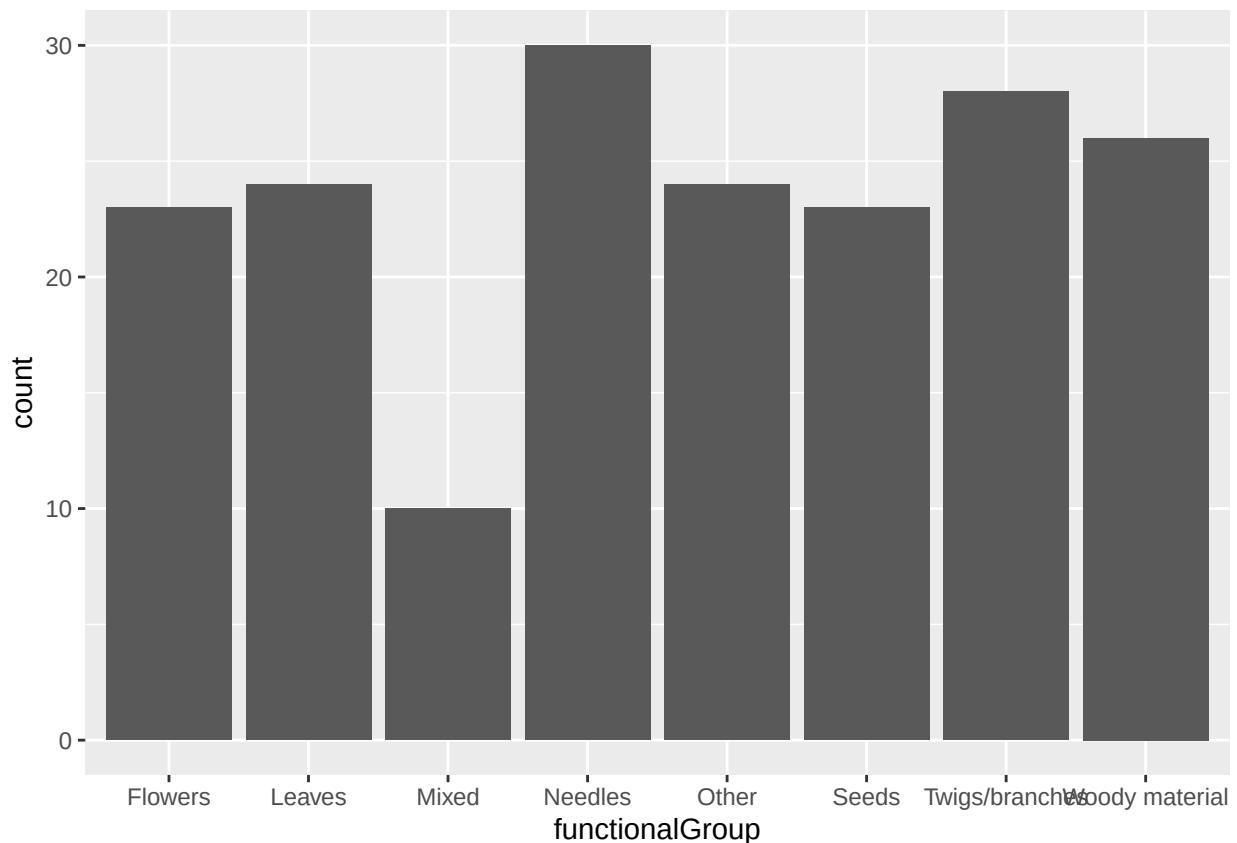
```
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
##      20      19      18      15      14       8      16      17
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
##      14      14      16      17
```

Answer: The ‘unique’ function just tells the viewer unique values within plotIDs as well as “Levels”, which reports the number of said unique values (in this case, 12). The ‘summary’ command displays both IDs and their frequency of occurrence within the data.

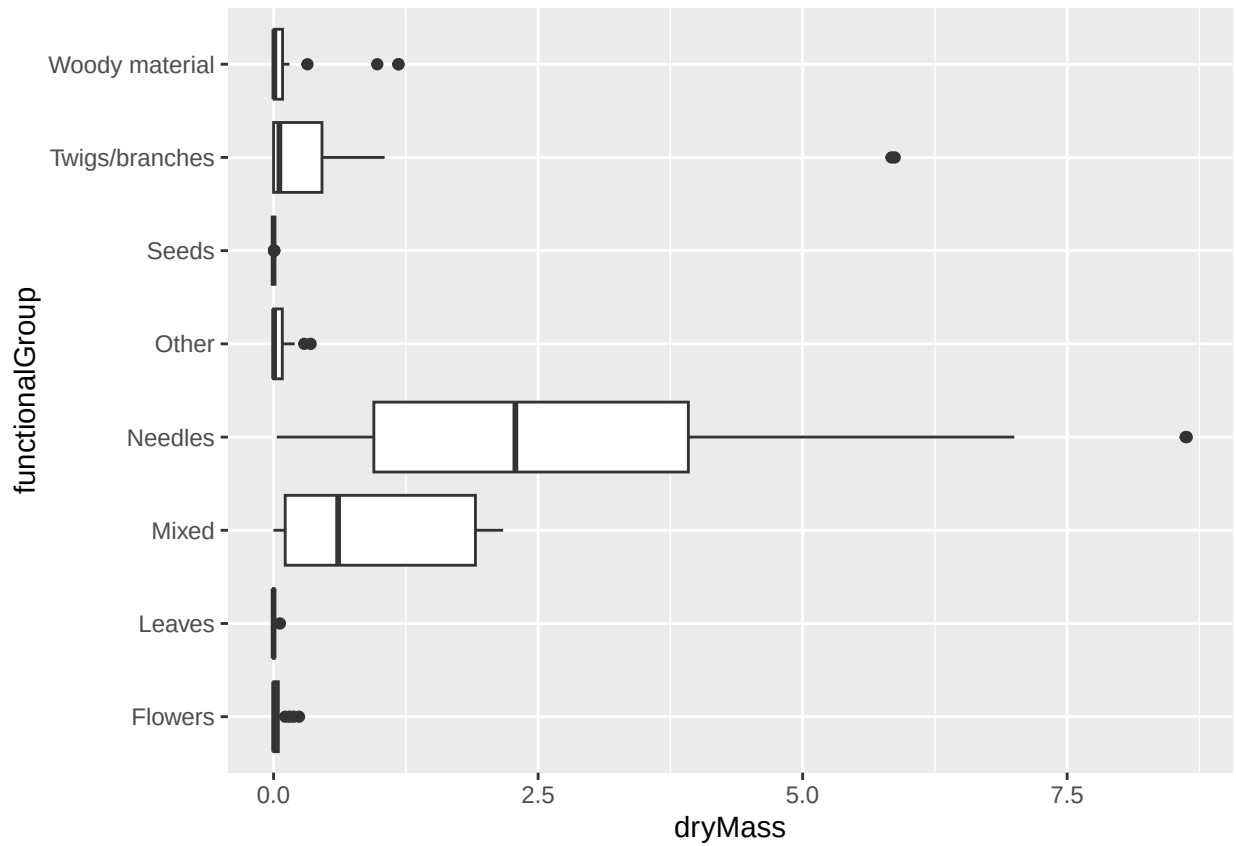
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot( data = Litter,
        aes(functionalGroup)) +
  geom_bar()
```

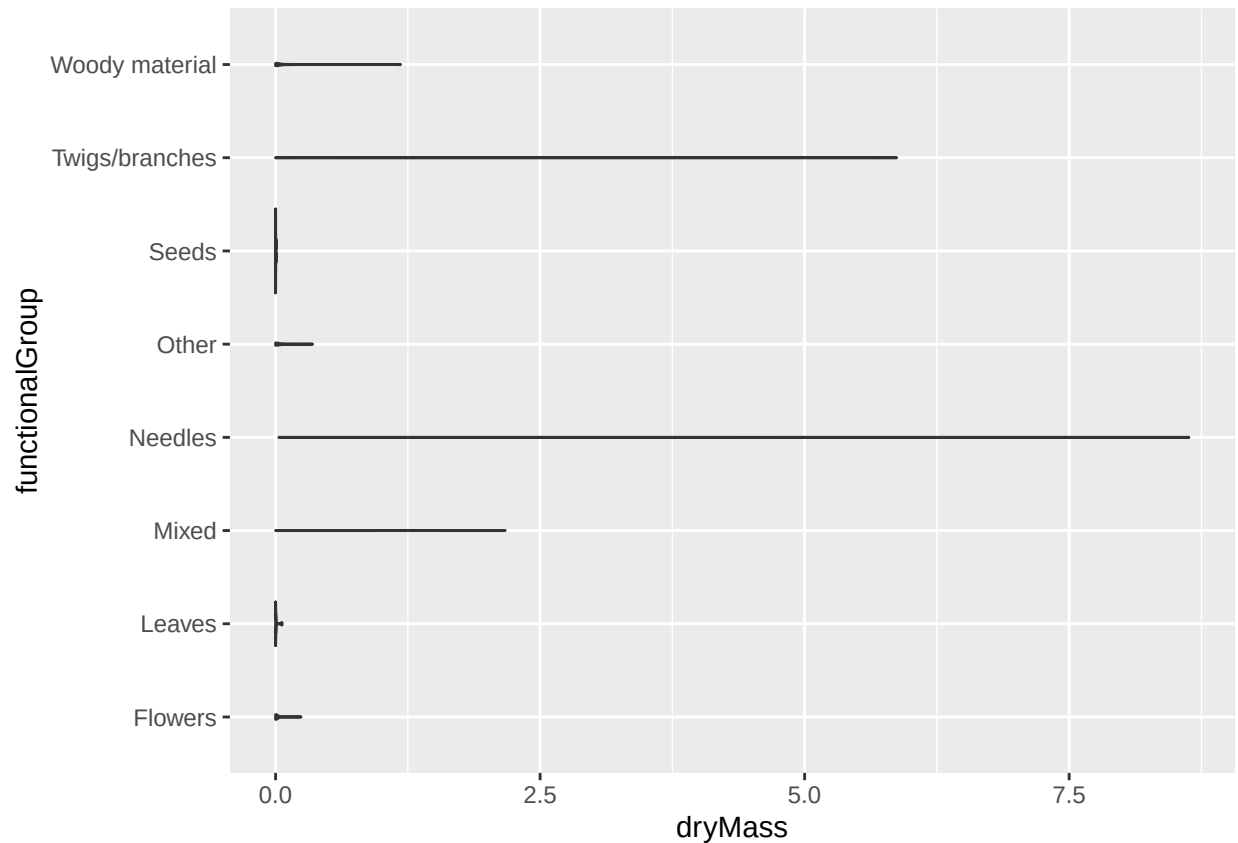


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(data = Litter,  
       aes(dryMass, functionalGroup))+  
geom_boxplot()
```



```
ggplot(data = Litter,  
       aes(dryMass, functionalGroup))+  
geom_violin()
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: The boxplot is a more effective visualization option than a violin plot because box plots show where data is most concentrated, the median of functional group dry mass, as well as outliers. The violin plot doesn't display density of values, thus potentially misrepresenting where the most values lie. If we relied solely on the violin plot, we would have mistakenly found that twigs/branches are the second highest biomass source at the sites due to a few outliers.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Overall, needles tend to have the highest biomass, followed by mixed litter.